Control Systems - Lecture Notes

1. System Modelling

A linear time-invariant system is described by a transfer function G(s) = Y(s)/U(s) evaluated with zero initial conditions, where s is the Laplace variable. The order of the system is the degree of the denominator polynomial. The poles are the roots of the denominator and govern the transient response; the zeros are the roots of the numerator.

A first-order system is written G(s) = K/(1 + tau*s), where K is the DC gain and tau is the time constant. Its step response reaches 63.2 percent of the final value at t = tau, and settles to within 2 percent in about 4*tau.

2. Second-Order Systems

The standard second-order form is G(s) = omega_n^2 / (s^2 + 2*zeta*omega_n*s + omega_n^2), where omega_n is the natural frequency and zeta is the damping ratio. For zeta < 1 the system is underdamped and the peak overshoot is Mp = exp(-pi*zeta / sqrt(1 - zeta^2)). For zeta = 1 it is critically damped, and for zeta > 1 overdamped.

The 2 percent settling time is approximately Ts = 4/(zeta*omega_n). The damped natural frequency is omega_d = omega_n*sqrt(1 - zeta^2).

3. Stability

A system is stable if all poles lie in the left half of the s-plane (negative real part). The Routh-Hurwitz criterion determines stability from the coefficients of the characteristic polynomial without solving for the roots. The number of sign changes in the first column of the Routh array equals the number of right-half-plane poles.

A marginally stable or unstable system has poles on the imaginary axis or in the right half plane respectively.

4. PID Control

A PID controller produces u(t) = Kp*e(t) + Ki*integral of e dtau + Kd*de/dt. The proportional term Kp reduces rise time but leaves steady-state error; the integral term Ki removes steady-state error but can increase overshoot; the derivative term Kd reduces overshoot and improves settling time but amplifies noise.

The Ziegler-Nichols method tunes gains from the ultimate gain Ku and ultimate period Tu: for a PID controller Kp = 0.6*Ku, Ti = 0.5*Tu and Td = 0.125*Tu.